GlycoTwin

Personalized Post-Meal Glucose Digital Twin

Digital Twin Challenge 2026 · Happiest Health Summit 2026

Team: *[Team name / member names — to be filled in]*
Institution: *[College / university — to be filled in]*
Incubator (if any): *[Incubator name, or "Not applicable" — to be filled in]*

Problem statement

After the same meal, two people can show very different post-meal glucose responses — age, body composition, activity and individual metabolism all matter. Most glucose-forecasting tools give one prediction and move on: they don't remember the person, don't say how confident they are, and don't improve as more is learned about that individual.

WHY IT MATTERS

Post-meal glucose excursions are a well-studied correlate of metabolic risk. A tool that is honest about uncertainty and adapts to the individual is a more defensible research direction than a single population-average number presented with false confidence.

Healthcare use case

WHO

A research/self-monitoring context: an individual (or a researcher studying a cohort) wearing a continuous glucose monitor who wants to understand how their own glucose responds to specific meals and activity — **not** a clinical decision-support tool.

WHAT THE TOOL DOES

For a planned meal, forecasts the probability and range of the 2-hour glucose maximum, personalized to that individual's own history, and updates itself as outcomes are observed.

EXPLICITLY OUT OF SCOPE

Diagnosis, insulin dosing, medication or diet prescriptions, and any clinical decision. GlycoTwin is an investigational research prototype — every page of the application and this deck says so.

Proposed solution

THE LOOP

Forecast → Observe → Update. A Digital Twin holds a personal, versioned state. It forecasts a meal's glucose outcome with uncertainty, receives the real outcome once observed, and — if that outcome is eligible — runs a closed-form Bayesian update that writes a new, numbered twin version. The next forecast uses the updated state.

1 · FORECAST

Probability + 90% interval for the 2-hour glucose maximum

2 · OBSERVE

Real outcome recorded once the window closes

3 · UPDATE

Eligible outcomes create a new, auditable twin version

Innovative features

1 · PERSISTENT, VERSIONED TWIN

Each eligible outcome creates a new version. Nothing is overwritten; full audit history.

2 · UNCERTAINTY-AWARE FORECASTING

A distribution for the 2-hour maximum, a 90% interval, and P(max ≥ 180 mg/dL) — never a bare point estimate.

3 · SEQUENTIAL BAYESIAN PERSONALIZATION

Model B's carbohydrate sensitivity updates in closed form from each person's own eligible meals — a parameter update, not retraining.

4 · EXPLAINABILITY + MODEL COMPARISON

Term-by-term explanations; Models B and C shown side by side with pre-declared comparison rules.

CGMacros dataset and research methodology

Participants	45, ~10 days of minute-level data each
Rows (audit)	687,580 total (5,655–18,735 per participant)
Signals	Two CGMs (Libre GL, Dexcom GL), activity (METs), heart rate, logged macros, meal photos
Target	Max glucose in (t0, t0+120 min] ≥ 180 mg/dL, where t0 = the logged meal row
Leakage control	Past-only features · participant-level folds · sequential (online) evaluation
Licence	CC BY-NC-SA 4.0 — cited, never redistributed, never committed to this repo

Source: data/audit/dataset_aggregate_summary.md, docs/data_validity_rules.md, docs/primary_channel_decision.md

Architecture

FRONTEND

React 19 + TypeScript + Vite · 7 pages · calls /api

BACKEND

FastAPI, 127.0.0.1 only · session + ownership on every twin route · scrypt hashing

DATABASE

SQLite, 11 tables (users, sessions, twins, twin_states, forecasts, observations…)

RESEARCH ENGINE (shared)

Bayesian Models B & C; the live twin and offline experiments call the exact same update/forecast functions.

OFFLINE CGMACROS PIPELINE

Runs on the researcher's own machine. Only aggregate priors and reports ever cross to the website.

Full diagram: docs/submission/GlycoTwin_Architecture.pdf

Models, uncertainty and personalization

$$\text{Model B: } \text{rise} = \beta \cdot \text{carbs} + \text{noise}$$

$$\text{Model C: } \text{rise} = (\beta + \gamma \cdot \text{activity}) \cdot \text{carbs} + \text{noise}$$

β starts from a population prior and is updated after each eligible meal in closed form (conjugate Normal update). The forecast's predictive variance combines noise and the twin's own parameter uncertainty — reported as a 90% interval and P(max ≥ 180 mg/dL), never a guarantee.

Eligible only: window closed, unusable windows excluded, back-dated forecasts refused. **A parameter update, not retraining a neural network.**

Forecast → Observe → Update, demonstrated live

	Twin version 0	Twin version 1 (after one eligible observation)
Model B β	0.747 ± 0.187	0.950 ± 0.144
Forecast (60g carbs, activity 0.3, glucose 110)	expected max 155, 90% interval 126–184	expected max 167, 90% interval 140–194
P(max ≥ 180 mg/dL)	7.7%	21.0%

Same inputs, different twin state, different forecast. This is what a one-off prediction tool cannot do.

Values entered by hand for this demonstration (60g carbs, 185 mg/dL outcome); not a participant's data. Full walkthrough: demo/video/GlycoTwin_demo_subtitled.mp4

Verified experimental results

Metric	Model B	Model C	C − B	95% interval
ROC-AUC	0.7943	0.8059	+0.0116	[−0.0084, +0.0479]
Brier score	0.1391	0.1379	−0.0012	[−0.0064, +0.0030]
Rise MAE (mg/dL)	29.40	29.51	+0.11	[−0.52, +0.68]

8 of 9 paired intervals contain zero → INCONCLUSIVE

963 meals, 34 participants, Libre channel, out-of-fold + sequential evaluation. Numbers are the researcher's recorded local runs (docs/model_b_vs_c_report.md), not recomputed for this deck. The shuffled-history / running-rate / population-XGBoost comparison is implemented but has no real-data run yet — no numbers are claimed for it.

Limitations, safety and future work

Investigational research prototype. Not a diagnostic tool. Not a validated medical device. Not for insulin dosing or treatment decisions. The current evaluation does not establish clinical effectiveness.

- 34 participants; the label is a lower bound (maximum *observed* glucose)
- Self-reported macronutrients; wrist-device activity; one dataset
- No multiplicity correction across the nine compared metrics
- No external security review of the account system; no Windows testing yet

NEXT

Run the full prespecified key experiment on real data with shuffled-history and baseline controls; test whether an activity effect depends on activity level.

Conclusion

GlycoTwin shows that a stateful, uncertainty-aware, Bayesian-personalized Digital Twin loop can be built and evaluated honestly end to end — and reports, without overstating it, that its first activity-conditioning hypothesis is currently **inconclusive**.

Repository: github.com/medhachoudhari/GlycoTwin
Demo video (15:34): *[Unlisted YouTube link — pending]*
Architecture: docs/submission/GlycoTwin_Architecture.pdf
License: *[pending your approval — see README]*